JASWANTH MADA

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Summary

Applied AI / Machine Learning Engineer specializing in multi-agent orchestration, production RAG pipelines, and LLM application development.

EXPERIENCE

Zenith Technology **Jul 2026 - Present**

AI/ML Engineer

- Architected an **agentic QA automation platform** using **Python, LangGraph,** and Playwright that converts user stories and acceptance criteria into structured test plans and executable browser workflows, leading to **50% reduction** in manual test preparation time.
- Built independent **test execution and verification agents** that evaluate UI behavior using DOM state, screenshots, network responses, console logs, and expected acceptance criteria, with a 100-case human-reviewed evaluation set designed to measure pass/fail agreement.
- Engineered an automated **failure investigation and recovery workflow** that distinguishes application defects, agent execution errors, and infrastructure failures, gathers supporting browser and application evidence, retries recoverable failures.
- Implemented **stage-specific context and evidence selection**, retry policies, and telemetry to track token usage, test latency, agent failures, and model cost, leading to a **25% reduction in LLM token consumption** without degrading verification quality.

Accessifiers **Aug 2025 - Jun 2026**

Machine Learning Engineer

- Engineered a RAG document intelligence platform using **Python, FastAPI, LangGraph, LangChain and Pinecone**, enabling research and engineering teams to query enterprise policy and technical documentation with source-grounded responses and reducing document lookup time by **35%**.
- Built an event-driven **AWS S3 + Lambda ingestion pipeline** to parse, chunk, embed **100K+ pages** of PDF and Markdown content, maintaining an up-to-date vector knowledge base while avoiding unnecessary full-document reprocessing.
- Implemented **hybrid retrieval, metadata filtering, reranking, and iterative retrieval workflows**, leading to higher answer relevance and reduced irrelevant answers by **25%.**
- Developed an automated **LLM evaluation framework** using RAGAS, pytest, and GitHub Actions to measure retrieval quality and answer groundedness before deployment, reducing release-related issues by **30%**.
- Containerized the **FastAPI** inference service with **Docker** and logged response latency, token consumption, retrieval failures, application errors, and model costs, maintaining average end-to-end response latency below **3 seconds**.

Skills

- **Programming & Core ML**: Python, NumPy, Pandas, Scikit Learn, PyTorch
- **Agentic AI & Orchestration**: Crew AI, Multi Agent Systems, ReAct Pattern, Planning Pattern, Tool Use, Agent Memory Management, Guardrails & Output Validation, Human-In-The-Loop, Agent Callbacks
- **Cloud & AWS**: EC2, S3, Lambda, API Gateway, SQS, DynamoDB, ElasticCache(Redis), CloudWatch
- **Databases**: MySQL, PostgreSQL, DynamoDB, Redis
- **Software Engineering**: OOP, Data Structures & Algorithms, Design Patterns, Multithreading, Concurrency
- **LLMs & Retrieval Systems**: RAG, LangChain, Pinecone, Prompt Engineering, Embedding Based Search

PROJECTS

Job Application Agent | Python, LangGraph, LangChain, MCP, Playwright, SQLite, Apify Github

- End-to-end orchestration: Engineered a LangGraph workflow that processes job queues, validates opportunities, navigates employer portals, handles authentication, and submits applications.
- Personalized application materials: Generates role-specific LaTeX/PDF resumes and drafts screening-question answers using candidate experience, project evidence, and job requirements.
- Reliable browser automation: Integrated MCP browser tools with form-field verification and repair to handle dynamic application forms and detect unsuccessful inputs.
- Human-in-the-loop recovery: Pauses for CAPTCHAs, security challenges, and unanswered questions, with persistent profile memory that reuses answers at the appropriate candidate, employer, or job scope.
- Resumable execution: Uses SQLite checkpoints, structured model outputs, retry policies, and diagnostic logging to support recovery and troubleshooting across application sessions.

EDUCATION

Purdue University Northwest **Aug 2023 - May 2025**

Masters, Computer Science *Hammond, IN*

Achievements

- **ICPC:**Ranked 43 out of 89 teams in the USA Mid-Center Regional ICPC Contest.